Skill 11: React API Integration - Fetching Data Using Fetch API, Axios & Local JSON

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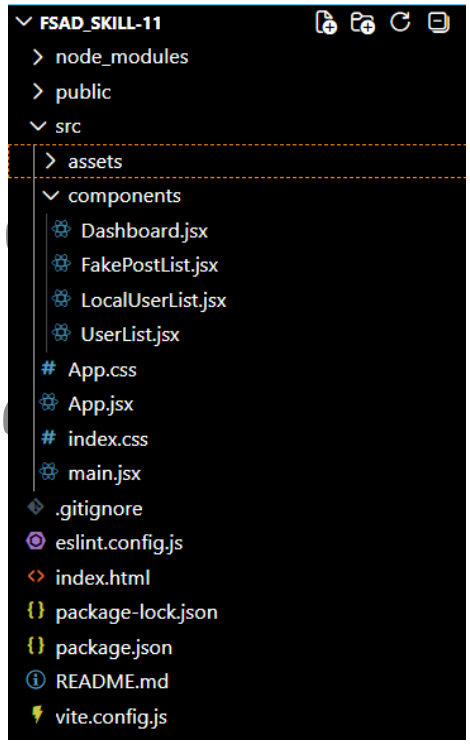
Section: 2

Scenario:

A news portal wants to fetch and display user and article information dynamically using React. Developers must learn how useState, useEffect, fetch(), axios, and local JSON work. This Skill demonstrates how frontend applications interact with local data, public APIs, and fake APIs, and how different components are navigated using hyperlinks.

Structure:



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Codes:

Dashboard.jsx:

```
import React, { useState } from "react";

import LocalUserList from "./LocalUserList";

import UserList from "./UserList";

import FakePostList from "./FakePostList";


function Dashboard() {
```

```
  const [page, setPage] = useState("");

  return (
    <div>
      <h1>News Portal Dashboard</h1>

      <button onClick={() => setPage("local")}>Local Users</button>
      <button onClick={() => setPage("api")}>Users API</button>
      <button onClick={() => setPage("posts")}>Fake API Posts</button>

      <hr />

      {page === "local" && <LocalUserList />}
      {page === "api" && <UserList />}
      {page === "posts" && <FakePostList />}
    </div>
  );
}

export default Dashboard;
```

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FakePostList.jsx:

```
import React, { useEffect, useState } from "react";
import axios from "axios";

function FakePostList() {
  const [posts, setPosts] = useState([]);
  const [filtered, setFiltered] = useState([]);
  const [userId, setUserId] = useState("");

  const fetchData = () => {
```

```
  axios.get("https://dummyjson.com/posts")
    .then((res) => {
      setPosts(res.data.posts);
      setFiltered(res.data.posts);
    })
    .catch(() => {
      alert("Error fetching posts");
    });
};

useEffect(() => {
  fetchData();
}, []);

const handleFilter = (id) => {
  setUserId(id);
  if (id === "") {
    setFiltered(posts);
  } else {
    setFiltered(posts.filter(post => post.userId === parseInt(id)));
  }
};

return (
  <div>
    <h2>Fake API Posts</h2>

    <button onClick={fetchData}>Refresh</button>

    <select value={userId} onChange={(e) => handleFilter(e.target.value)}>
      <option value="">All Users</option>
```

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```
      <option value="1">User 1</option>
      <option value="2">User 2</option>
      <option value="3">User 3</option>
    </select>

    {filtered.map(post => (
      <div key={post.id} className="card">
        <h4>{post.title}</h4>
        <p>{post.body}</p>
      </div>
    ))}
  </div>
);
}

export default FakePostList;
```

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LocalUserList.jsx:

```
import React, { useEffect, useState } from "react";

function LocalUserList() {
  const [users, setUsers] = useState([]);
  const [loading, setLoading] = useState(true);
  const [error, setError] = useState("");

  useEffect(() => {
    fetch("/users.json")
      .then((res) => {
        if (!res.ok) throw new Error("Failed to fetch local data");
        return res.json();
      })
```

```
    .then((data) => {
      setUsers(data);
      setLoading(false);
    })
    .catch((err) => {
      setError(err.message);
      setLoading(false);
    });
  }, []);

  if (loading) return <p>Loading local users...</p>;
  if (error) return <p>{error}</p>;

  return (
    <div>
      <h2>Local Users</h2>
      {users.map((user) => (
        <div key={user.id} className="card">
          <p><b>{user.name}</b></p>
          <p>{user.email}</p>
          <p>{user.phone}</p>
        </div>
      ))}
    </div>
  );
}

export default LocalUserList;
```

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UserList.jsx:

```
import React, { useEffect, useState } from "react";
```

```
function UserList() {
  const [users, setUsers] = useState([]);
  const [loading, setLoading] = useState(true);
  const [error, setError] = useState("");

  useEffect(() => {
    fetch("https://jsonplaceholder.typicode.com/users")
      .then((res) => {
        if (!res.ok) throw new Error("API Error");
        return res.json();
      })
      .then((data) => {
        setUsers(data);
        setLoading(false);
      })
      .catch((err) => {
        setError(err.message);
        setLoading(false);
      });
  }, []);

  if (loading) return <p>Loading API users...</p>;
  if (error) return <p>{error}</p>;

  return (
    <div>
      <h2>Users from API</h2>
      {users.map((user) => (
        <div key={user.id} className="card">
          <p><b>{user.name}</b></p>
```

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```
        <p>{user.email}</p>
        <p>{user.phone}</p>
      </div>
    ))}
  </div>
 );
}

export default UserList;
```

App.css:

```
body {
  font-family: Arial;
  background-color: #f5f5f5;
}

.app {
  text-align: center;
}

button {
  margin: 10px;
  padding: 10px 15px;
  cursor: pointer;
}

.card {
  background: white;
  padding: 10px;
  margin: 10px auto;
  width: 300px;
```

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```
  border-radius: 8px;
  box-shadow: 0 0 5px gray;
}
```

App.jsx:

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <link rel="icon" type="image/svg+xml" href="/favicon.svg" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>fsad_skill-11</title>
  </head>
  <body>
    <div id="root"></div>
    <script type="module" src="/src/main.jsx"></script>
  </body>
</html>
```

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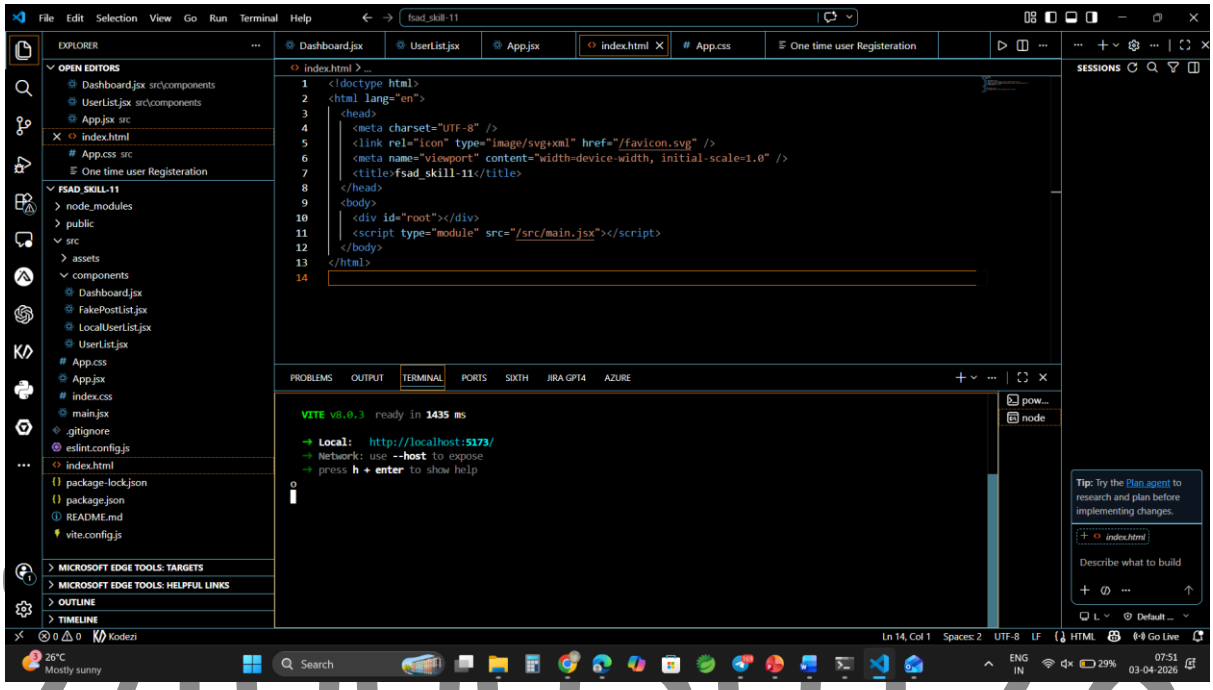
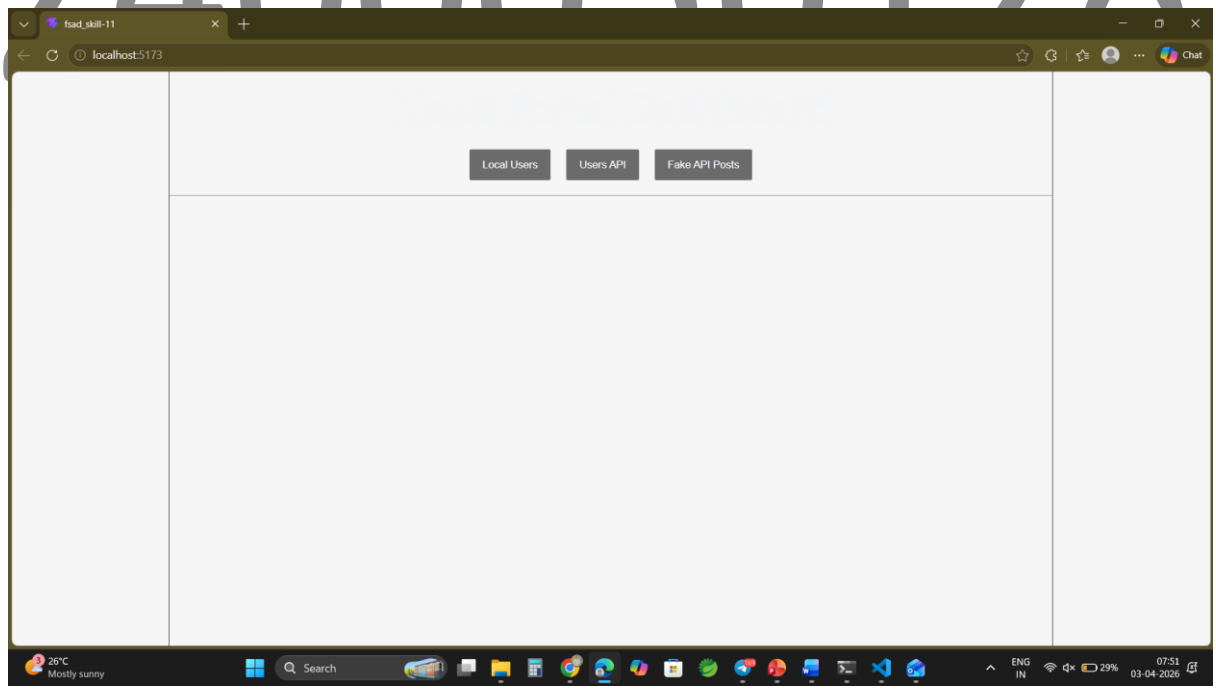
Index.html:

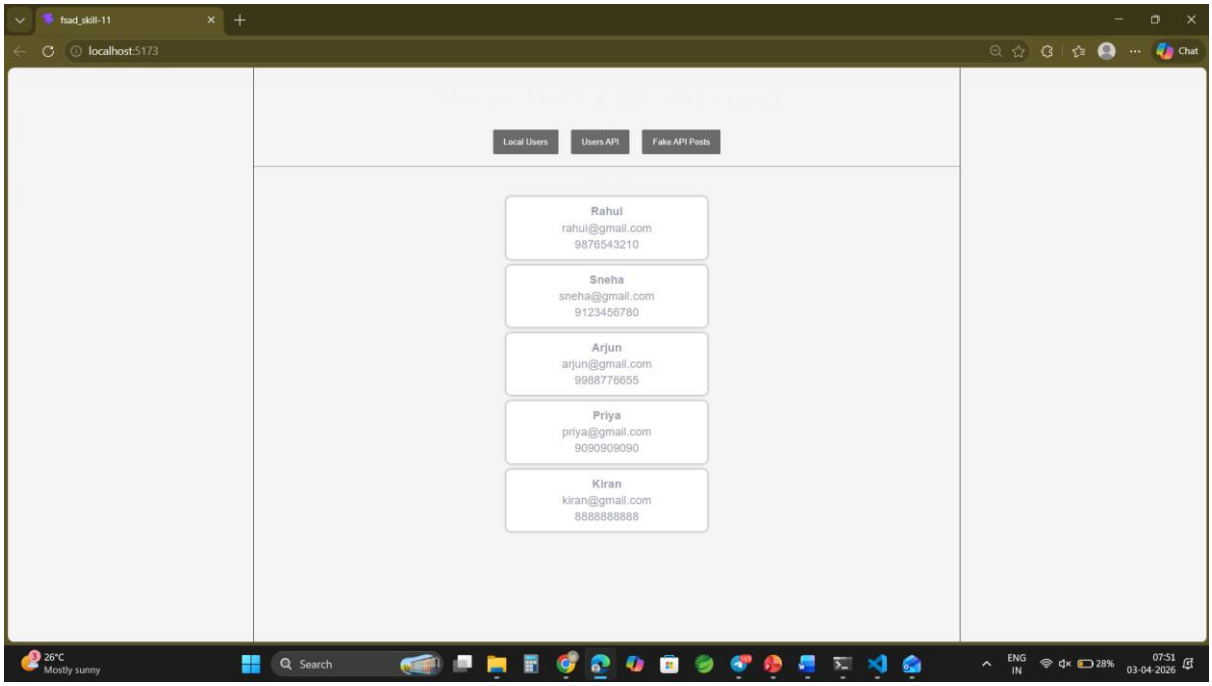
```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <link rel="icon" type="image/svg+xml" href="/favicon.svg" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>fsad_skill-11</title>
  </head>
  <body>
    <div id="root">
    <script type="module" src="/src/main.jsx"></script>
```

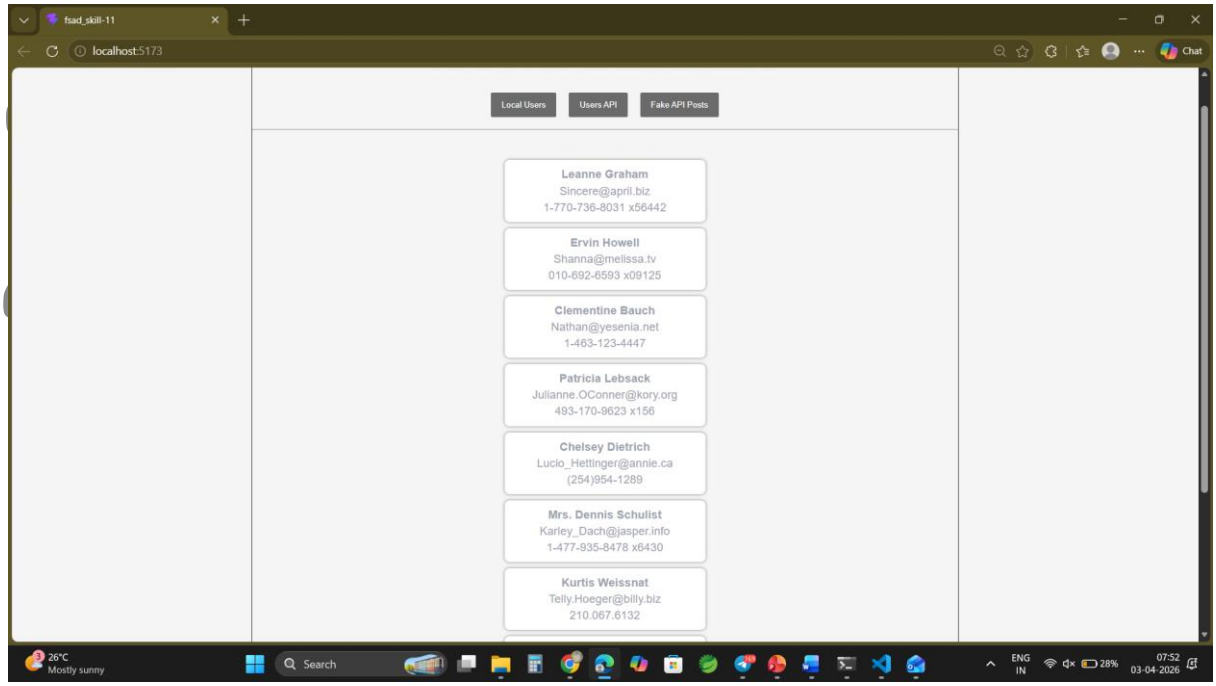
```
</body>
</html>
```

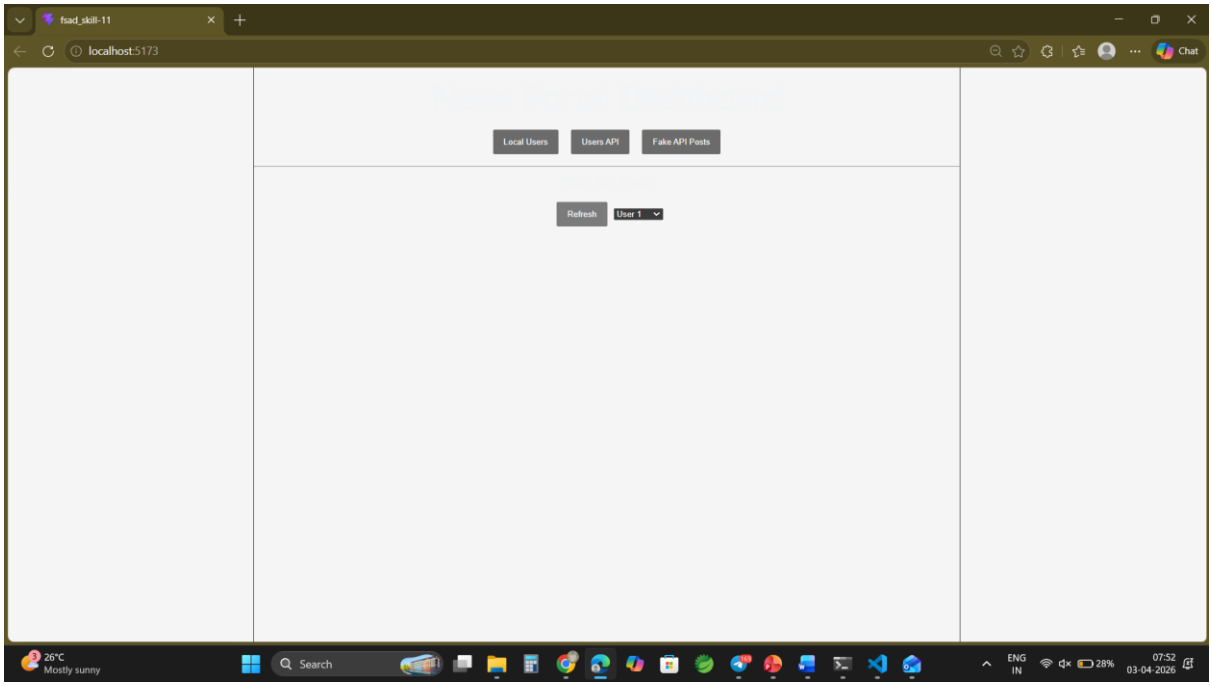
OUTPUTS:






fsad_skill-11
localhost:5173
Chat
Local Users
Users API
Fake API Posts
Rahul
rahul@gmail.com
9876543210
Sneha
sneha@gmail.com
9123456780
Arjun
arjun@gmail.com
9988776655
Priya
priya@gmail.com
9090909090
Kiran
kiran@gmail.com
8888888888
26°C
Mostly sunny
Search
ENG
IN
28%
07:51
03-04-2026


fsad_skill-11
localhost:5173
Chat
Local Users
Users API
Fake API Posts
Leanne Graham
Sincere@april.biz
1-770-736-8031 x56442
Ervin Howell
Shanna@melissa.tv
010-692-6593 x09125
Clementine Bauch
Nathan@yesenia.net
1-463-123-4447
Patricia Lebsack
Julianne.OConner@kory.org
493-170-9623 x156
Chelsey Dietrich
Lucio_Hettinger@annie.ca
(254)954-1289
Mrs. Dennis Schulist
Karley_Dach@jasper.info
1-477-935-8478 x6430
Kurtis Weissnat
Telly.Hoeger@billy.biz
210.067.6132
26°C
Mostly sunny
Search
ENG
IN
28%
07:52
03-04-2026


fsad_skill-11
localhost:5173
Chat
Local Users
Users API
Fake API Posts
Refresh
User 1
26°C
Mostly sunny
Search
ENG
IN
28%
07:52
03-04-2026

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